

# Sorghum downy mildew

*Peronosclerospora sorghi*



Symptoms on sorghum leaf  
(Rob Williams)



Symptoms on sorghum leaves  
(Rob Williams)



Symptoms on sorghum heads  
(Rob Williams)

## Prevention

- Use resistant varieties if available.
- Use clean, healthy seed.
- Plant crop early, when there is a lower prevalence of downy mildew.
- Plant when soil temperature is below 20°C which is unfavorable for downy mildew development.
- Alternate sorghum with crops other than maize, which may be infected by the same strains of downy mildew. Crop rotation will limit build-up of the disease in the soil.
- Remove weeds, particularly related weed hosts such as Columbus grass (*Sorghum alnum*) to reduce potential sources of the disease.
- Seeds for sowing next season should be dried to less than 20% moisture to reduce the risk of disease transmission.
- Keep crop density as low as possible to increase aeration.
- If using non-resistant varieties, minimise application of high levels of nitrogen to make plants less susceptible to attack.

## Monitoring

- Additional relevant crops: maize, pearl millet.
- Check crop at least every two weeks.
- Leaves (not all symptoms may show):
- New leaves first show a loss of colour (chlorosis) before developing white stripes, widening with subsequent leaf production – this differs from much narrower streaks associated with maize dwarf mosaic virus.
- Narrower and more erect, compared to leaves of healthy plants. The white areas eventually dry up, going reddish-brown. Leaves eventually shred.
- Brown, rectangular lesions on leaves. Downy or woolly appearance on underside of leaves.
- Plant:
- Stunted and usually does not produce any grain.
- Monitor crop closely when humidity and temperature are high as downy mildew favours these conditions.

## Direct Control

- Remove infected crop debris if possible and burn or deep plough (at least to 20 cm depth).
- Minimise movement of people and livestock in affected areas so as not to spread the pathogen.

Note: Pesticides may be available to control this pest. Please check with the Ministry of Agriculture in your country to find out which pesticides are registered in your country and the local restrictions for their use.